

Target Company Brazil Region (SQL Project)

Project Overview

This project focuses on analyzing a Brazilian e-commerce dataset using SQL to uncover insights related to customer behavior, sales growth, delivery performance, freight costs, payment methods, and regional order distribution.

The analysis was performed using multiple SQL concepts including:

- Joins
- Common Table Expressions (CTEs)
- Aggregate Functions
- Date & Time Functions
- Window Functions
- Conditional Statements
- Grouping & Sorting

The dataset contains information related to:

- Customers
 - Orders
 - Payments
 - Order Items
 - Geolocation
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1. Exploratory Data Analysis (EDA)

Objectives

- Understand dataset structure
- Check available tables and columns
- Analyze order timeline
- Identify customer locations

Key Analysis Performed

- Previewed records from customers and geolocation tables

- Identified the overall order purchase period using MIN and MAX timestamps
- Extracted customer cities and states for orders placed during Jan–Mar 2018

Key Insights

- Orders span across multiple years indicating long-term operational data
 - Customers belong to various cities and states across Brazil
 - Regional order analysis helps identify active customer locations
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2. In-Depth Exploration

2.1 Order Growth Trend

Objective

Analyze whether the number of orders increased over the years.

Analysis

Grouped orders by purchase year and counted total orders.

Insight

- Orders showed significant year-over-year growth since 2016
 - Indicates expanding e-commerce adoption and business growth
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2.2 Monthly Seasonality Analysis

Objective

Identify seasonal ordering patterns across months.

Analysis

Counted total orders for each month.

Insight

- August and May recorded the highest order volumes
- October and November showed relatively lower performance

- Remaining months demonstrated moderate sales activity
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2.3 Customer Ordering Time Analysis

Objective

Determine during which part of the day customers place most orders.

Time Categories

- Dawn: 0–6 hrs
- Morning: 7–12 hrs
- Afternoon: 13–18 hrs
- Night: 19–23 hrs

Insight

- Most orders were placed during the Afternoon
 - Night orders were the second highest
 - Dawn recorded the lowest activity
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3. Evolution of E-Commerce Orders Across Brazil

3.1 Month-on-Month Orders by State

Objective

Analyze monthly order trends for each Brazilian state.

Analysis

Grouped orders by:

- State
- Year
- Month

Insight

- Different states showed varying order growth patterns
 - Some states consistently generated higher order volumes
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3.2 Customer Distribution Across States

Objective

Identify states with the highest and lowest customer counts.

Analysis

Counted distinct customers per state.

Insight

- São Paulo (SP) had the highest customer base
 - Roraima (RR) had the lowest customer count
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4. Economic Impact Analysis

4.1 Percentage Increase in Order Cost (2017–2018)

Objective

Calculate growth in total order payments between 2017 and 2018 (Jan–Aug).

Techniques Used

- CTEs
- Window Functions (LEAD)

Insight

- Total order cost increased by more than 100%
 - Reflects strong revenue growth
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4.2 Total & Average Order Price by State

Objective

Analyze state-wise order value performance.

Insight

- São Paulo (SP) generated the highest total sales value
 - Paraíba (PB) showed the highest average order value
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4.3 Total & Average Freight Value by State

Objective

Evaluate freight expenses across states.

Insight

- Freight costs varied significantly by state
 - Some states showed high average logistics expenses
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5. Sales, Freight & Delivery Performance Analysis

5.1 Delivery Time Analysis

Objective

Calculate:

- Actual delivery time
- Difference between estimated and actual delivery dates

Metrics

- Time to Deliver
- Estimated vs Actual Delivery Difference

Insight

- Enables identification of delayed and on-time deliveries

- Helps evaluate logistics efficiency
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5.2 Top 5 States with Highest & Lowest Freight Costs

Objective

Compare freight costs across states.

Insight

- Freight value fluctuations vary significantly between states
 - Useful for logistics optimization strategies
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5.3 Top States by Average Delivery Time

Objective

Find states with the highest average delivery duration.

Insight

- Delivery times differed considerably across regions
 - Some states experienced much longer shipping durations
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6. Payment Analysis

6.1 Monthly Orders by Payment Type

Objective

Analyze payment method preferences over time.

Payment Types Analyzed

- Credit Card
- Debit Card
- Voucher
- Boleto

- Others

Insight

- Certain payment methods were more widely preferred
 - Trends can help improve payment strategy and customer experience
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6.2 Orders Based on Installments

Objective

Analyze order counts by payment installment usage.

Insight

- A large number of orders were paid in a single installment
 - Indicates strong preference for one-time payments
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SQL Concepts Used

- SELECT Statements
 - INNER JOIN
 - Aggregate Functions
 - COUNT()
 - SUM()
 - AVG()
 - ROUND()
 - Date Functions
 - EXTRACT()
 - DATE_DIFF()
 - Conditional Logic
 - CASE WHEN
 - Window Functions
 - LEAD()
 - Common Table Expressions (CTEs)
 - GROUP BY & ORDER BY
 - UNION ALL
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Overall Business Insights

1. The Brazilian e-commerce market experienced strong yearly growth.
 2. Afternoon and night are the most active order periods.
 3. São Paulo dominates in terms of customer base and total sales value.
 4. Freight and delivery performance vary significantly across states.
 5. Payment installment behavior suggests many customers prefer single-payment transactions.
 6. Seasonal trends indicate peak ordering months that can support future marketing strategies.
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Conclusion

This project demonstrates how SQL can be used to perform comprehensive business analysis on e-commerce datasets. The insights generated can support:

- Business decision-making
- Logistics optimization
- Customer behavior analysis
- Revenue growth strategies
- Regional performance evaluation

The project also highlights practical applications of SQL in real-world data analytics scenarios.